Machine Learning Operation

Reflection and Further Learning

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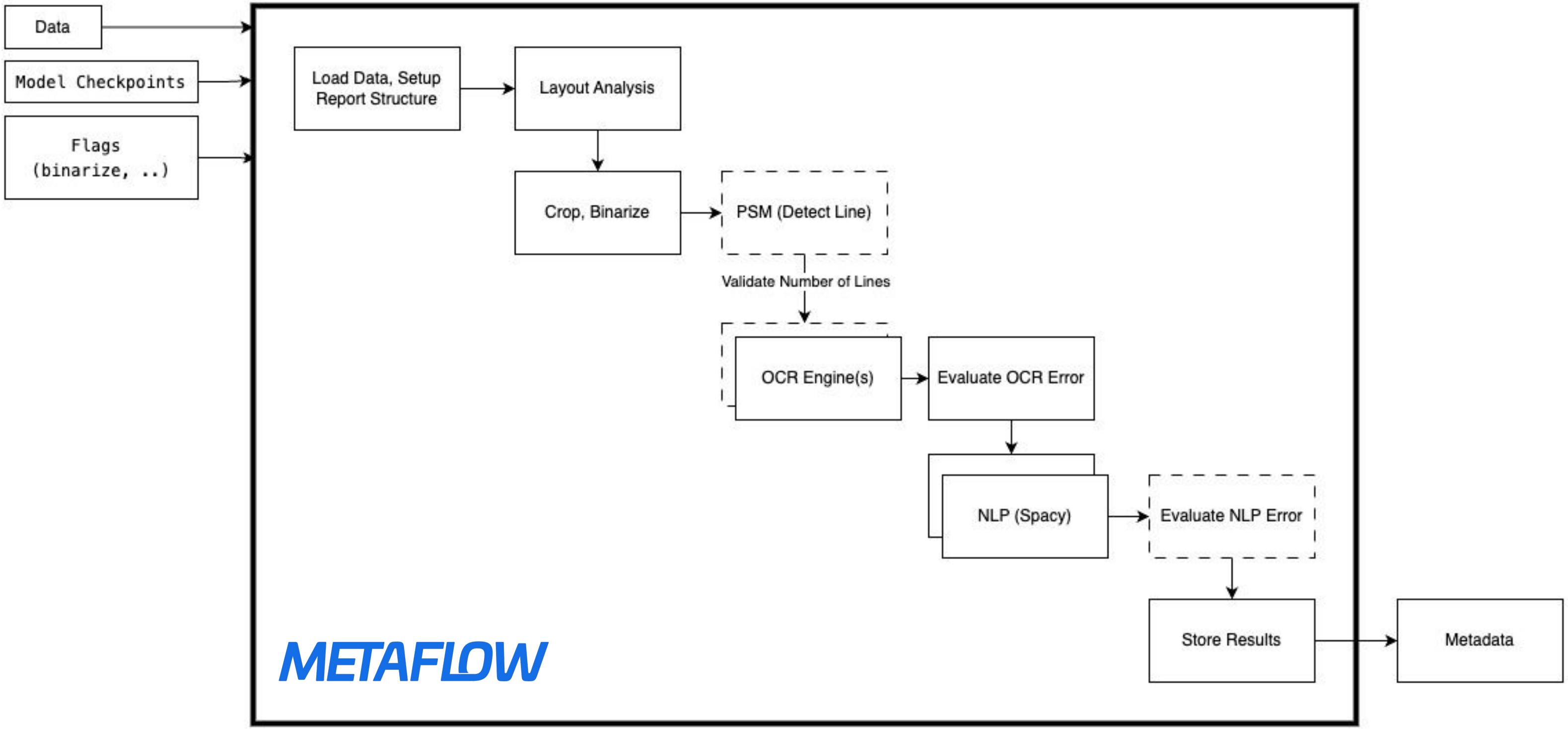
Tasks (Data Validation, Pipeline, Training, Model Registry)

Decision	Question	Choices	Details	
data validations	what parameters I should validate in my data and how?	consistency(null, range and is in list), completeness or others	consistency	since for this problem, checking the data schema was enough, but it depends on the expectation of model performance
data validations	where to run my tests?	in a Dockerfile volume or in a second process	in a second process	I would go with a copy of data in a volume attached to a container (scalable, isolation)
pipeline	what tools and why?	metaflow or kubeflow	metaflow	metaflow since it was simpler, however KFP also supports local pipelines
robustness	how I find out the result of current training is robust?	compare it with the best model r2 metric or define a raw baseline	compare with best model	depends on project definition, I’m not sure what are the future consequences of this choice

Tasks (Drift Detection, Pipeline Version, A/B Test)

Decision	Question	Choices	Details	
drift detection	what part of my solution i need to observe at this stage?	features importance, data drift, labels	input data drift	since my problem is in early stage of feature engineering, the drift in inputs is important for me, my model is a decision tree based on vector of inputs, when they are drifted i may need to retrain my model
A/B test	how to do it?	pass model object, its uri, run id, model uri	i have passed the model uri	user i could use the model name i used during training and the version, the rest is taken care of by flow itself
Experiment test	how do i test when i have more than one model?	pass all of them or pass the reference to an experiment	pass an experiment	user only need to specify the experiment name and the flow handles the rest
Consistency	how to do have the same preprocessing for data?	save the proprocess job	save the proprocess job	i faced multiple issues, since the features were being different on my new input, so i saved the preprocessed as an artifact in the model and used it during evaluation

Apply same mindset to Document Analysis Project (Failure)



Reflection

confidence in Basic ML (Decision Trees, SVN, Boosters)	how to increase the confidence in a training job?	what is an example of a multi-model inference pipeline?
metadata produced by Metaflow got very large, I don't know the reason	How does a reproducible project look like?	how to estimate such a project delivery? what is SLA? Costs?
I think tests / quality gates are not enough	different team in mid-large companies, what about small team where there are only two practitioners?	do I even need a pipeline and framework? or I'm expected to implement one in a work setup?
Deep Learning, LLMs and other types of ML algorithms, training jobs	I don't have any sense on very large batch pipelines, what could be a very large such system?	Implement a Pipeline for DRL Agent

Further Learning

Internship backed by CNCF - Volcano.sh
(Linux Foundation)

Enhancement of JobFlow functionalities

- introduce parameter patching in JobFlow
- work on fault tolerance, implement a few mechanism (backOff Policy, maxRetry, retry Policy)
- control flow statements (if/else, switch, for)

Internship backed by Google - Kubeflow
(Google Summer of Code)

Implement JAX and TF Runtimes for KF Trainer V2

- based on JobSet API
- distributed training on K8s
- blueprints
- Python SDK